

1.实验目的

- (1) 理解汇编语言源程序的基本组成结构与核心要素；
- (2) 掌握汇编语言程序从编写、编译、链接到运行的完整流程，熟悉相关操作环境与基本步骤；
- (3) 自主学习并掌握使用 DEBUG 命令进行程序调试的常用方法，能独立完成基础调试操作。

2.实验环境

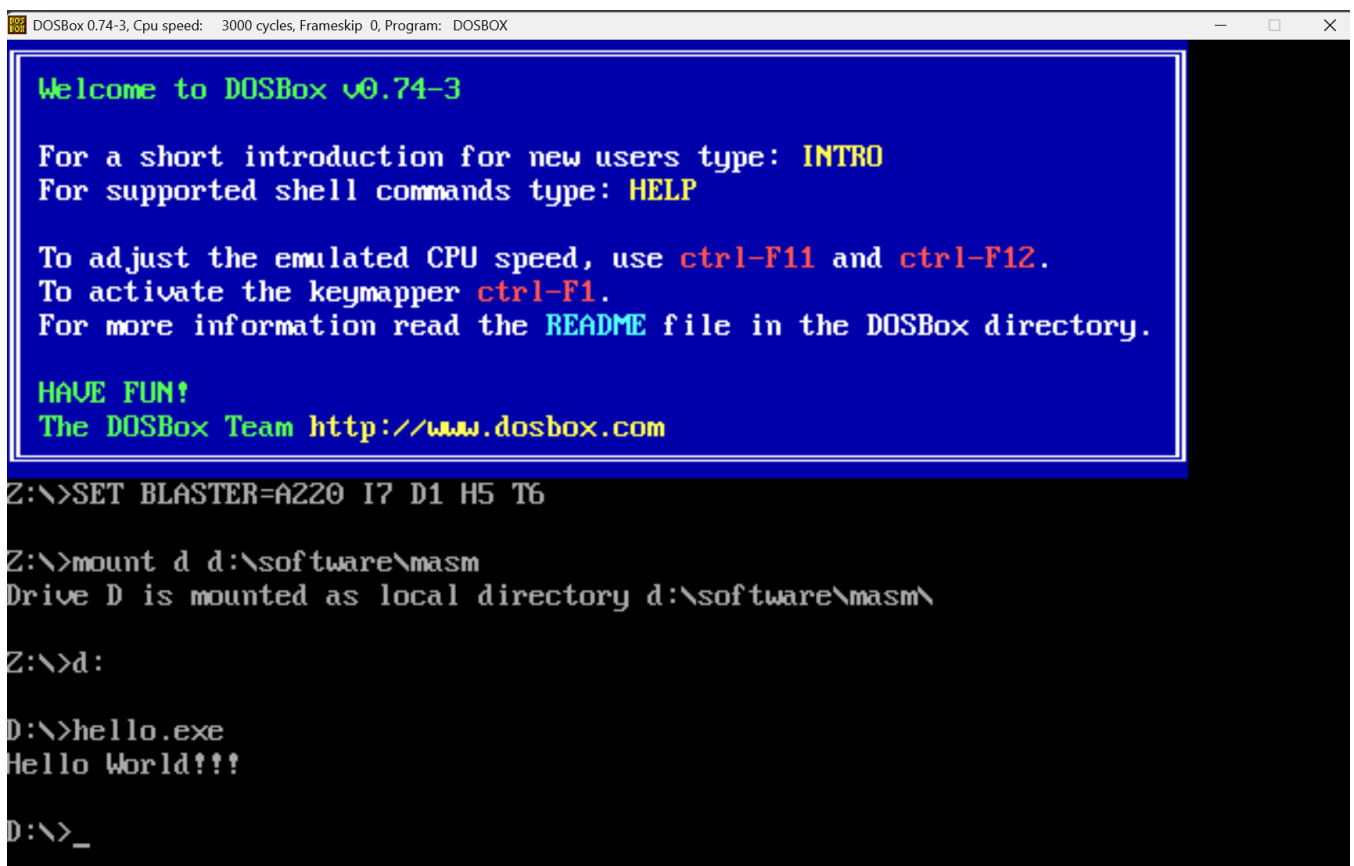
通过 DOSBox 模拟器，在 64 位操作系统中搭建 DOS 环境，作为汇编语言程序的开发与运行平台。

3.实验内容

- (1) 针对给定的两个例程（例程 a、例程 b），分别通过“DOS 环境模拟”与“集成开发环境”两种方式，完成程序的编辑、汇编、链接及调试全流程操作；
- (2) 提前预习汇编语言常用助记词的含义与使用规则，熟悉基本汇编操作逻辑，同时掌握 Debug 调试程序的核心操作方法。

4.实验具体实现

(1) 环境配置完成后就可以正常执行hello.exe了：



```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DOSBOX

Welcome to DOSBox v0.74-3

For a short introduction for new users type: INTRO
For supported shell commands type: HELP

To adjust the emulated CPU speed, use ctrl-F11 and ctrl-F12.
To activate the keymapper ctrl-F1.
For more information read the README file in the DOSBox directory.

HAVE FUN!
The DOSBox Team http://www.dosbox.com

Z:\>SET BLASTER=A220 I7 D1 H5 T6

Z:\>mount d d:\software\masm
Drive D is mounted as local directory d:\software\masm\

Z:\>d:

D:\>hello.exe
Hello World!!!

D:\>_
```

(2) 编写程序：

先创建后缀为 asm 的文件，将给出的例程分别在上面复刻出来，具体如下图：

文件(F) 编辑(E) 选择(S) 查看(V) 转到(G) 运行(R) 终端(T) ... < >

ASM hello.ASM ASM hw.ASM X

D: > Software > MASM > ASM hw.ASM

```
1 data segment
2     string DB 'Hello World!!!',0AH,0DH,'$'
3 data ends
4 code segment
5     assume cs:code, ds:data
6 start:
7     MOV AX,DATA
8     MOV DS,AX
9
10    mov dx,OFFSET string
11    mov AH,09H
12    int 21H
13
14    MOV AH,4CH
15    int 21H
16 code ends
17 end start
```

(3) 对例程进行编译和链接及运行：

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DOSBOX
D:\>hello.exe
Hello World!!!

D:\>ml /c hw.asm
Microsoft (R) Macro Assembler Version 6.11
Copyright (C) Microsoft Corp 1981-1993. All rights reserved.

Assembling: hw.asm

D:\>link hw.obj

Microsoft (R) Segmented Executable Linker Version 5.60.339 Dec 5 1994
Copyright (C) Microsoft Corp 1984-1993. All rights reserved.

Run File [hw.exe]:
List File [nul.map]:
Libraries [.lib]:
Definitions File [nul.def]:
LINK : warning L4021: no stack segment

D:\>hw.exe
Hello World!!!

D:\>
```

解释：这段汇编代码的功能是在 DOS 环境下输出字符串 “Hello World!!!” 并换行，然后退出程序。

首先这里定义了一个名为 `string` 的字节变量，存储的内容是字符串 'Hello world!!!'，以及两个控制字符：0AH（换行符）和 0DH（回车符），最后以 '\$' 结尾。'\$' 是字符串的结束标记，用于指示输出到此处停止。

然后初始化数据段寄存器 `DS`，使其指向定义字符串的 `data` 段；调用 DOS 中断的 09H 功能输出指定字符串；最后调用 4CH 功能退出程序。

运行后会在 DOS 界面输出 “Hello World!!!” 并换行，然后程序退出。

在这里把输出的内容修改为 I study program software engineering in xMU 并输出, 测试程序:

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DOSBOX
D:\>hw.exe
Hello World!!!

D:\>ml /c hw.asm
Microsoft (R) Macro Assembler Version 6.11
Copyright (C) Microsoft Corp 1981-1993. All rights reserved.

Assembling: hw.asm

D:\>link hw.obj

Microsoft (R) Segmented Executable Linker Version 5.60.339 Dec 5 1994
Copyright (C) Microsoft Corp 1984-1993. All rights reserved.

Run File [hw.exe]:
List File [nul.map]:
Libraries [.lib]:
Definitions File [nul.def]:
LINK : warning L4021: no stack segment

D:\>hw.exe
I study program software engineering in xMU

D:\>
```

(4) 熟悉debug操作并利用debug调试程序

a. 查看寄存器和反汇编程序 (r、u)

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
D:\>hw.exe
I study program software engineering in xMU

D:\>debug hw.exe
-r
AX=FFFF BX=0000 CX=0040 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=075A ES=075A SS=0769 CS=076D IP=0000  NU UP EI PL NZ NA PO NC
076D:0000 B86A07      MOV     AX,076A
-u
076D:0000 B86A07      MOV     AX,076A
076D:0003 8ED8      MOV     DS,AX
076D:0005 BA0000      MOV     DX,0000
076D:0008 B409      MOV     AH,09
076D:000A CD21      INT     21
076D:000C B44C      MOV     AH,4C
076D:000E CD21      INT     21
076D:0010 56        PUSH    SI
076D:0011 1AB80400  SBB     BH,[BX+SI+0004]
076D:0015 50        PUSH    AX
076D:0016 0E        PUSH    CS
076D:0017 E8A60A      CALL    0AC0
076D:001A B88127      MOV     AX,2781
076D:001D 50        PUSH    AX
076D:001E FF761A      PUSH    [BP+1A]
_
```

b.查看dump (d)

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
076D:0036 741B JZ 0053
076D:0038 B80800 MOV AX,0008
076D:003B 50 PUSH AX
076D:003C 0E PUSH CS
076D:003D E8800A CALL 0AC0
076D:0040 FF36FA56 PUSH [56FA]
-d
075A:0180 83 C4 08 B8 04 00 50 0E-E8 65 0A B8 1C 27 50 FF .....P..e...P.
075A:0190 76 FE FF 76 FC B8 0A 53-50 0E E8 5D 08 83 C4 0A v..v...SP..l....
075A:01A0 83 3E 12 58 02 7D 6E 8B-46 F8 8B 56 FA 03 46 F4 .>.X.}n.F..U..F.
075A:01B0 13 56 F6 03 46 F0 13 56-F2 03 46 DE 13 56 E0 03 .U..F..U..F..U..
075A:01C0 46 08 13 56 0A 03 46 0C-13 56 0E 03 46 10 13 56 F..U..F..U..F..U
075A:01D0 12 03 46 14 13 56 16 03-46 DA 13 56 DC 03 46 EC ..F..U..F..U..F.
075A:01E0 13 56 EE 03 46 E8 13 56-EA 03 46 D6 13 56 D8 89 .U..F..U..F..U..
075A:01F0 46 D2 89 56 D4 B9 04 00-51 0E E8 F3 09 B8 7F 27 F..U....Q.....'
-d 0100
075A:0100 49 20 73 74 75 64 79 20-70 72 6F 67 72 61 6D 20 I study program
075A:0110 73 6F 66 74 77 61 72 65-20 65 6E 67 69 6E 65 65 software enginee
075A:0120 72 69 6E 67 20 69 6E 20-58 4D 55 0A 0D 24 00 00 ring in XMU..$.
075A:0130 B8 6A 07 8E D8 BA 00 00-B4 09 CD 21 B4 4C CD 21 .j.....!.L.!
075A:0140 56 1A B8 04 00 50 0E E8-A6 0A B8 81 27 50 FF 76 U....P.....'P.v
075A:0150 1A FF 76 18 B8 F2 52 50-0E E8 9E 08 83 C4 0A A1 ..v...RP.....
075A:0160 F8 56 0B 06 FA 56 74 1B-B8 08 00 50 0E E8 80 0A .U...Ut....P....
075A:0170 FF 36 FA 56 FF 36 F8 56-B8 FA 52 50 0E E8 7A 08 .6.U.6.U..RP..z.
```

c.全速运行和退出 (g、q)

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DOSBOX
-d
075A:0180 83 C4 08 B8 04 00 50 0E-E8 65 0A B8 1C 27 50 FF .....P..e...P.
075A:0190 76 FE FF 76 FC B8 0A 53-50 0E E8 5D 08 83 C4 0A v..v...SP..l....
075A:01A0 83 3E 12 58 02 7D 6E 8B-46 F8 8B 56 FA 03 46 F4 .>.X.}n.F..U..F.
075A:01B0 13 56 F6 03 46 F0 13 56-F2 03 46 DE 13 56 E0 03 .U..F..U..F..U..
075A:01C0 46 08 13 56 0A 03 46 0C-13 56 0E 03 46 10 13 56 F..U..F..U..F..U
075A:01D0 12 03 46 14 13 56 16 03-46 DA 13 56 DC 03 46 EC ..F..U..F..U..F.
075A:01E0 13 56 EE 03 46 E8 13 56-EA 03 46 D6 13 56 D8 89 .U..F..U..F..U..
075A:01F0 46 D2 89 56 D4 B9 04 00-51 0E E8 F3 09 B8 7F 27 F..U....Q.....'
-d 0100
075A:0100 49 20 73 74 75 64 79 20-70 72 6F 67 72 61 6D 20 I study program
075A:0110 73 6F 66 74 77 61 72 65-20 65 6E 67 69 6E 65 65 software enginee
075A:0120 72 69 6E 67 20 69 6E 20-58 4D 55 0A 0D 24 00 00 ring in XMU..$.
075A:0130 B8 6A 07 8E D8 BA 00 00-B4 09 CD 21 B4 4C CD 21 .j.....!.L.!
075A:0140 56 1A B8 04 00 50 0E E8-A6 0A B8 81 27 50 FF 76 U....P.....'P.v
075A:0150 1A FF 76 18 B8 F2 52 50-0E E8 9E 08 83 C4 0A A1 ..v...RP.....
075A:0160 F8 56 0B 06 FA 56 74 1B-B8 08 00 50 0E E8 80 0A .U...Ut....P....
075A:0170 FF 36 FA 56 FF 36 F8 56-B8 FA 52 50 0E E8 7A 08 .6.U.6.U..RP..z.
-g
I study program software engineering in XMU
Program terminated normally
-q
D:\>
```

d.查看dump并且修改内容 (d、e)

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
-d 0100
075A:0100  49 20 73 74 75 64 79 20-70 72 6F 67 72 61 6D 20  I study program
075A:0110  73 6F 66 74 77 61 72 65-20 65 6E 67 69 6E 65 65  software enginee
075A:0120  72 69 6E 67 20 69 6E 20-58 4D 55 0A 0D 24 00 00  ring in XMU..$.
075A:0130  B8 6A 07 8E D8 BA 00 00-B4 09 CD 21 B4 4C CD 21  .j.....!.L.!
075A:0140  56 1A B8 04 00 50 0E E8-A6 0A B8 81 27 50 FF 76  U....P.....'P.
075A:0150  1A FF 76 18 B8 F2 52 50-0E E8 9E 08 83 C4 0A A1  ..v...RP.....
075A:0160  F8 56 0B 06 FA 56 74 1B-B8 08 00 50 0E E8 80 0A  .U...Ut....P....
075A:0170  FF 36 FA 56 FF 36 F8 56-B8 FA 52 50 0E E8 7A 08  .6.U.6.U..RP..z.
-e 0100
075A:0100  49.69

-e 0110
075A:0110  73.53

-d 0100
075A:0100  69 20 73 74 75 64 79 20-70 72 6F 67 72 61 6D 20  i study program
075A:0110  53 6F 66 74 77 61 72 65-20 65 6E 67 69 6E 65 65  Software enginee
075A:0120  72 69 6E 67 20 69 6E 20-58 4D 55 0A 0D 24 00 00  ring in XMU..$.
075A:0130  B8 6A 07 8E D8 BA 00 00-B4 09 CD 21 B4 4C CD 21  .j.....!.L.!
075A:0140  56 1A B8 04 00 50 0E E8-A6 0A B8 81 27 50 FF 76  U....P.....'P.
075A:0150  1A FF 76 18 B8 F2 52 50-0E E8 9E 08 83 C4 0A A1  ..v...RP.....
075A:0160  F8 56 0B 06 FA 56 74 1B-B8 08 00 50 0E E8 80 0A  .U...Ut....P....
075A:0170  FF 36 FA 56 FF 36 F8 56-B8 FA 52 50 0E E8 7A 08  .6.U.6.U..RP..z.
_
```

e.跟踪指令 (t、p)

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
AX=076A BX=0000 CX=0040 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=075A ES=075A SS=0769 CS=076D IP=0003  NU UP EI PL NZ NA PO NC
076D:0003 8ED8          MOV     DS,AX

AX=076A BX=0000 CX=0040 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=076D IP=0005  NU UP EI PL NZ NA PO NC
076D:0005 BA0000     MOV     DX,0000

AX=076A BX=0000 CX=0040 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=076D IP=0008  NU UP EI PL NZ NA PO NC
076D:0008 B409          MOV     AH,09

AX=096A BX=0000 CX=0040 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=076D IP=000A  NU UP EI PL NZ NA PO NC
076D:000A CD21          INT     21

AX=096A BX=0000 CX=0040 DX=0000 SP=FFFA BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=F000 IP=14A0  NU UP DI PL NZ NA PO NC
F000:14A0 FB          STI

AX=096A BX=0000 CX=0040 DX=0000 SP=FFFA BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=F000 IP=14A1  NU UP EI PL NZ NA PO NC
F000:14A1 FE3B          ???     [BX+SI]          DS:0000=49
_
```

```

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
DS=076A ES=075A SS=0769 CS=076D IP=0005  NU UP EI PL NZ NA PO NC
076D:0005 BA0000          MOV     DX,0000

AX=076A BX=0000 CX=0040 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=076D IP=0008  NU UP EI PL NZ NA PO NC
076D:0008 B409          MOV     AH,09

AX=096A BX=0000 CX=0040 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=076D IP=000A  NU UP EI PL NZ NA PO NC
076D:000A CD21          INT     21

AX=096A BX=0000 CX=0040 DX=0000 SP=FFFA BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=F000 IP=14A0  NU UP DI PL NZ NA PO NC
F000:14A0 FB          STI

AX=096A BX=0000 CX=0040 DX=0000 SP=FFFA BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=F000 IP=14A1  NU UP EI PL NZ NA PO NC
F000:14A1 FE38          ???     [BX+SI]          DS:0000=49
-t
I study program software engineering in XMU

AX=096A BX=0000 CX=0040 DX=0000 SP=FFFA BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=F000 IP=14A5  NU UP EI PL NZ NA PO NC
F000:14A5 CF          IRET

```

```

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
F000:14A0 FB          STI
-p

AX=096A BX=0000 CX=0040 DX=0000 SP=FFFA BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=F000 IP=14A1  NU UP EI PL NZ NA PO NC
F000:14A1 FE38          ???     [BX+SI]          DS:0000=49
-p
I study program software engineering in XMU

AX=096A BX=0000 CX=0040 DX=0000 SP=FFFA BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=F000 IP=14A5  NU UP EI PL NZ NA PO NC
F000:14A5 CF          IRET
-p

AX=096A BX=0000 CX=0040 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=076D IP=000C  NU UP EI PL NZ NA PO NC
076D:000C B44C          MOV     AH,4C
-p 2

AX=4C6A BX=0000 CX=0040 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=076A ES=075A SS=0769 CS=076D IP=000E  NU UP EI PL NZ NA PO NC
076D:000E CD21          INT     21

Program terminated normally

```

f.覆写指令 (a)

先查看反汇编结果：


```

076D:0016 0E          PUSH    CS
076D:0017 E8A60A      CALL    0AC0
076D:001A B88127      MOV     AX,2781
076D:001D 50          PUSH    AX
076D:001E FF761A      PUSH    [BP+1A]

```

-q

D:\>debug hw.exe

-u

```

076D:0000 B86A07      MOV     AX,076A
076D:0003 8ED8      MOV     DS,AX
076D:0005 BA0000      MOV     DX,0000
076D:0008 B409      MOV     AH,09
076D:000A CD21      INT     21
076D:000C B44C      MOV     AH,4C
076D:000E CD21      INT     21
076D:0010 CC      INT     3
076D:0011 1AB80400    SBB     BH,[BX+SI+0004]
076D:0015 50          PUSH    AX
076D:0016 0E          PUSH    CS
076D:0017 E8A60A      CALL    0AC0
076D:001A B88127      MOV     AX,2781
076D:001D 50          PUSH    AX
076D:001E FF761A      PUSH    [BP+1A]

```

然后覆写指令：


```
076D:0016 0E      PUSH    CS
076D:0017 E8A60A    CALL    0AC0
076D:001A B88127    MOV     AX,2781
076D:001D 50      PUSH    AX
076D:001E FF761A    PUSH    [BP+1A]
-a 0000
076D:0000 MOV AX,0000
076D:0003
-u 0000
076D:0000 B80000    MOV     AX,0000
076D:0003 8ED8     MOV     DS,AX
076D:0005 BA0000    MOV     DX,0000
076D:0008 B409     MOV     AH,09
076D:000A CD21     INT     21
076D:000C B44C     MOV     AH,4C
076D:000E CD21     INT     21
076D:0010 CC      INT     3
076D:0011 1AB80400 SBB     BH,[BX+SI+0004]
076D:0015 50      PUSH    AX
076D:0016 0E      PUSH    CS
076D:0017 E8A60A    CALL    0AC0
076D:001A B88127    MOV     AX,2781
076D:001D 50      PUSH    AX
076D:001E FF761A    PUSH    [BP+1A]
```

g.比较指令及运算 (c, h)

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
-C DS:0100 DS:0110
  ^ Error
-C ds:0100 L 1 ds:0110
075A:0100 49 73 075A:0110
-q

D:\>debug hw.exe
-rds
DS 075A
:
-c 075A:0100 L 1 075A:0100
-c 075A:0110 L 1 075A:0100
075A:0110 73 49 075A:0100
-d 0100
075A:0100 49 20 73 74 75 64 79 20-70 72 6F 67 72 61 6D 20 I study program
075A:0110 73 6F 66 74 77 61 72 65-20 65 6E 67 69 6E 65 65 software enginee
075A:0120 72 69 6E 67 20 69 6E 20-58 4D 55 0A 0D 24 00 00 ring in XMU..$.
075A:0130 B8 6A 07 8E D8 BA 00 00-B4 09 CD 21 B4 4C CD 21 .j.....!.L.!
075A:0140 CC 1A B8 04 00 50 0E E8-A6 0A B8 81 27 50 FF 76 .....P.....'P.v
075A:0150 1A FF 76 18 B8 F2 52 50-0E E8 9E 08 83 C4 0A A1 ..v...RP.....
075A:0160 F8 56 0B 06 FA 56 74 1B-B8 08 00 50 0E E8 80 0A .U...Ut....P....
075A:0170 FF 36 FA 56 FF 36 F8 56-B8 FA 52 50 0E E8 7A 08 .6.U.6.U..RP..z.
-h 0100 0110
0210 FFF0
```

(5) 将"W"改为"k"

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
075A:0160 F8 56 0B 06 FA 56 74 1B-B8 08 00 50 0E E8 80 0A .U...Ut....P....
075A:0170 FF 36 FA 56 FF 36 F8 56-B8 FA 52 50 0E E8 7A 08 .6.U.6.U..RP..z.
-q

D:\>debug hello.exe
-d 0100
075A:0100 48 65 6C 6C 6F 20 57 6F-72 6C 64 21 21 21 0A 0D Hello World!!!!..
075A:0110 24 00 00 00 00 00 00 00-00 00 00 00 00 00 00 $.
075A:0120 B8 6A 07 8E D8 BA 00 00-B4 09 CD 21 B4 4C CD 21 .j.....!.L.!
075A:0130 B8 6A 07 8E D8 BA 00 00-B4 09 CD 21 B4 4C CD 21 .j.....!.L.!
075A:0140 CC 1A B8 04 00 50 0E E8-A6 0A B8 81 27 50 FF 76 .....P.....'P.v
075A:0150 1A FF 76 18 B8 F2 52 50-0E E8 9E 08 83 C4 0A A1 ..v...RP.....
075A:0160 F8 56 0B 06 FA 56 74 1B-B8 08 00 50 0E E8 80 0A .U...Ut....P....
075A:0170 FF 36 FA 56 FF 36 F8 56-B8 FA 52 50 0E E8 7A 08 .6.U.6.U..RP..z.
-e 0100 6B
-d 0100
075A:0100 6B 65 6C 6C 6F 20 57 6F-72 6C 64 21 21 21 0A 0D kello World!!!!..
075A:0110 24 00 00 00 00 00 00 00-00 00 00 00 00 00 00 $.
075A:0120 B8 6A 07 8E D8 BA 00 00-B4 09 CD 21 B4 4C CD 21 .j.....!.L.!
075A:0130 B8 6A 07 8E D8 BA 00 00-B4 09 CD 21 B4 4C CD 21 .j.....!.L.!
075A:0140 CC 1A B8 04 00 50 0E E8-A6 0A B8 81 27 50 FF 76 .....P.....'P.v
075A:0150 1A FF 76 18 B8 F2 52 50-0E E8 9E 08 83 C4 0A A1 ..v...RP.....
075A:0160 F8 56 0B 06 FA 56 74 1B-B8 08 00 50 0E E8 80 0A .U...Ut....P....
075A:0170 FF 36 FA 56 FF 36 F8 56-B8 FA 52 50 0E E8 7A 08 .6.U.6.U..RP..z.
```

5.实验分析与总结

通过这次实验，我初步了解了 dosbox 的使用和 debug 的使用，更深刻的理解了数据在寄存器中的存放。

了解了汇编的底层，熟悉了debug的指令和masm的相关指令